THE NOVA RESEARCH CURRICULUM

Answer key · Grade 7 · Term 2

For teachers only. Do not share with students.

Teacher answer key

English

Question bank

#	Outcome	Question	Answer
1	EN-7.2	"I am hungry," said Ali. Reported:	Ali said that I am hungry. / **Ali said that he was hungry.** / Ali says he hungry. / Ali said that he is hungry yesterday.
2	EN-7.2	In reported speech, "tomorrow" usually becomes …	yesterday / **the next day** / today / that week
3	EN-7.2	If you heat water to 100°C, it ___.	**boils** / would boil / will boiled / boiled
4	EN-7.2	If it rains, we ___ indoors.	play / **will play** / would played / played
5	EN-7.2	If I were a bird, I ___ fly.	will / **would** / am / can to
6	EN-7.3	A counter-argument is …	your main point / **the opposite view, which you then answer** / the title / a quotation only
7	EN-7.3	Write a position statement for "Homework does more good than harm".	Model: Homework helps students practise, plan their time and learn on their own, so it does more good than harm.
8	EN-7.3	An argumentative essay needs evidence for each point. True or false.	**True**

Urdu

Question bank

#	Outcome	Question	Answer
1	UR-7.2	«لب پہ آتی ہے دعا…» کس کا شعر ہے؟	غالب / **علامہ اقبال** / میر / حالی
2	UR-7.2	تشریح میں شامل ہوتا ہے …	صرف مشکل الفاظ / **حوالہ، مفہوم اور پیغام** / صرف شاعر کا نام / ایک نئی نظم
3	UR-7.3	«نو دو گیارہ ہونا» کا مطلب …	حساب کرنا / **بھاگ جانا** / خوش ہونا / سو جانا
4	UR-7.3	«آنکھوں کا تارا» کا مطلب …	ناراض / **بہت پیارا** / بیمار / دور

#	Outcome	Question	Answer
5	UR-7.3	«جیسی کرنی ویسی بھرنی» ضرب المثل ہے۔ True or false.	**True**
6	UR-7.3	«دال میں کالا ہونا» جملے میں استعمال کریں۔	Model: وہ بار بار بات بدل رہا تھا، ضرور دال میں کچھ کالا ہے۔

Mathematics

Question bank

#	Outcome	Question	Answer
1	MA-7.3	(x + 2)(x + 3) = ?	$x^2 + 6$ / **$x^2 + 5x + 6$** / $x^2 + 6x + 5$ / $2x + 5$
2	MA-7.3	$(a + b)^2$ = ?	$a^2 + b^2$ / **$a^2 + 2ab + b^2$** / $2a + 2b$ / $a^2 - b^2$
3	MA-7.3	Factorise $x^2 - 25$.	$(x - 5)^2$ / **$(x + 5)(x - 5)$** / $x(x - 25)$ / $(x + 25)(x - 1)$
4	MA-7.4	Solve $3(x - 2) = 12$.	$x = 2$ / **$x = 6$** / $x = 4$ / $x = 14/3$
5	MA-7.4	Solve $5x + 4 = 2x + 13$.	**$x = 3$** / $x = 17/7$ / $x = 9$ / $x = 4$
6	MA-7.4	For $y = 2x + 1$, what is y when x = 3?	5 / 6 / **7** / 9
7	MA-7.4	The point (0, 4) lies on the y-axis. True or false.	**True**
8	MA-7.5	Sum of interior angles of a hexagon?	540° / **720°** / 360° / 1,080°
9	MA-7.5	Exterior angles of any polygon add to …	180° / **360°** / 540° / it depends
10	MA-7.5	A circle has radius 7 cm. Its diameter is …	3.5 cm / **14 cm** / 49 cm / 21 cm

Physics

Question bank

#	Outcome	Question	Answer
1	PH-7.2	Sound cannot travel through …	water / steel / air / **a vacuum**
2	PH-7.2	Higher frequency gives a …	louder sound / **higher pitch** / slower sound / lower pitch
3	PH-7.2	Frequency is measured in …	metres / newtons / **hertz** / joules
4	PH-7.2	Sound is a longitudinal wave. True or false.	**True**
5	PH-7.3	Heat from the Sun reaches Earth by …	conduction / convection / **radiation** / evaporation
6	PH-7.3	A sea breeze is caused by …	conduction / **convection** / radiation / reflection

#	Outcome	Question	Answer
7	PH-7.3	25 °C in kelvin?	25 K / 248 K / **298 K** / 77 K
8	PH-7.3	100 °C in °F?	100 °F / 180 °F / **212 °F** / 373 °F
9	PH-7.3	Why are cooking pots often given plastic handles?	Model: plastic is an insulator, so heat is not conducted to your hand.

Chemistry

Question bank

#	Outcome	Question	Answer
1	CH-7.2	Which is a chemical change?	melting ice / dissolving salt / **burning paper** / cutting wood
2	CH-7.2	Melting chocolate is a physical change. True or false.	**True**
3	CH-7.2	Rusting needs iron, oxygen and …	salt / **water** / sunlight / carbon dioxide
4	CH-7.2	Galvanising coats iron with …	copper / gold / **zinc** / plastic
5	CH-7.2	The three sides of the fire triangle are fuel, heat and …	water / **oxygen** / nitrogen / smoke
6	CH-7.2	A fire blanket works by …	adding fuel / cooling only / **cutting off oxygen** / adding heat
7	CH-7.2	Give two signs that a chemical change has happened.	Model: a gas is given off and the colour changes (also heat, light, a precipitate).
8	CH-7.2	Burning fuel gives off carbon dioxide and water. True or false.	**True**

Biology

Question bank

#	Outcome	Question	Answer
1	BI-7.2	Gas exchange happens in the …	trachea / **alveoli** / heart / stomach
2	BI-7.2	How many chambers does the human heart have?	2 / 3 / **4** / 6
3	BI-7.2	Arteries carry blood … the heart.	to / **away from** / around / inside
4	BI-7.2	Which blood cells carry oxygen?	white cells / platelets / **red cells** / plasma
5	BI-7.2	Anaerobic respiration in muscles makes …	alcohol / **lactic acid** / oxygen / starch
6	BI-7.2	Breathing and respiration are the same thing. True or false.	**False**

#	Outcome	Question	Answer
7	BI-7.2	Why does your pulse rise during exercise?	Model: muscles need more oxygen and glucose, so the heart pumps faster.
8	BI-7.2	Yeast makes bread rise by releasing carbon dioxide. True or false.	**True**

History, Geography and Pakistan Studies

Question bank

#	Outcome	Question	Answer
1	HI-7.1	The court language of the Mughals was …	Arabic / Urdu / **Persian** / Turkish
2	HI-7.1	The Wazir Khan Mosque is in …	Karachi / **Lahore** / Multan / Peshawar
3	HI-7.2	Mughal decline began after Aurangzeb's death in …	1526 / 1657 / **1707** / 1857
4	HI-7.2	Nadir Shah invaded Delhi in 1739. True or false.	**True**
5	GE-7.1	Most monsoon rain in Pakistan falls in …	December–February / **July–September** / March–April / October
6	GE-7.2	Tarbela Dam is on the … River.	Jhelum / Ravi / **Indus** / Chenab
7	GE-7.2	The Indus Waters Treaty was signed in …	1947 / **1960** / 1973 / 1998
8	PS-7.2	Article 25-A gives children the right to …	vote / **free and compulsory education** / a job / travel abroad
9	PS-7.2	Name two duties of a citizen.	Model: obey the law and pay taxes (also: be loyal to the state, vote).

Islamiyat and Ethics

Question bank

#	Outcome	Question	Answer
1	IS-7.2	The Battle of the Trench took place in …	2 AH / 3 AH / **5 AH** / 8 AH
2	IS-7.2	Who suggested digging the trench?	Hazrat Umar (RA) / **Hazrat Salman Farsi (RA)** / Hazrat Ali (RA) / Hazrat Bilal (RA)
3	IS-7.2	The Treaty of Hudaybiyyah was made in …	4 AH / **6 AH** / 8 AH / 10 AH
4	IS-7.2	The treaty set a truce of …	one year / five years / **ten years** / forever
5	IS-7.2	The Qur'an called Hudaybiyyah a clear victory. True or false.	**True**

#	Outcome	Question	Answer
6	IS-7.2	What lesson does the Battle of the Trench teach?	Model: consult others and work hard together (also: patience, trust in Allah).
7	IS-7.4	The first caliph of Islam was …	Hazrat Umar (RA) / **Hazrat Abu Bakr (RA)** / Hazrat Uthman (RA) / Hazrat Ali (RA)
8	IS-7.4	Hazrat Khadijah (RA) was the first person to believe in the Prophet ﷺ. True or false.	**True**

Computer Science and skills

Question bank

#	Outcome	Question	Answer
1	CS-7.1	Which keyword starts a function in Python?	func / **def** / function / start
2	CS-7.1	What does `return` do?	prints text / **sends a value back from the function** / ends Python / makes a loop
3	CS-7.1	Write a function that doubles a number.	Model: `def double(n):` then `return n * 2`
4	CS-7.2	Which formula adds cells B2 to B10?	=ADD(B2:B10) / **=SUM(B2:B10)** / =B2+B10 / =TOTAL(B2-B10)
5	CS-7.2	Which formula finds the mean?	=MEAN() / =MID() / **=AVERAGE()** / =MODE() only
6	CS-7.2	A line chart is best for showing change over time. True or false.	**True**
7	RO-7.1	A motor is an … device.	input / **output** / storage / sensor
8	FL-7.1	Taxes help pay for public schools and roads. True or false.	**True**